🌐 Networking Notes

🖥️ Basics

Network

A **network** is a collection of computers connected to each other to share resources and data.

Internet

The **Internet** is a collection of interconnected computer networks.

Protocol

A **protocol** is a standardized set of rules that define how devices format, transmit, and receive data — acting as a common language for communication.

WWW (World Wide Web)

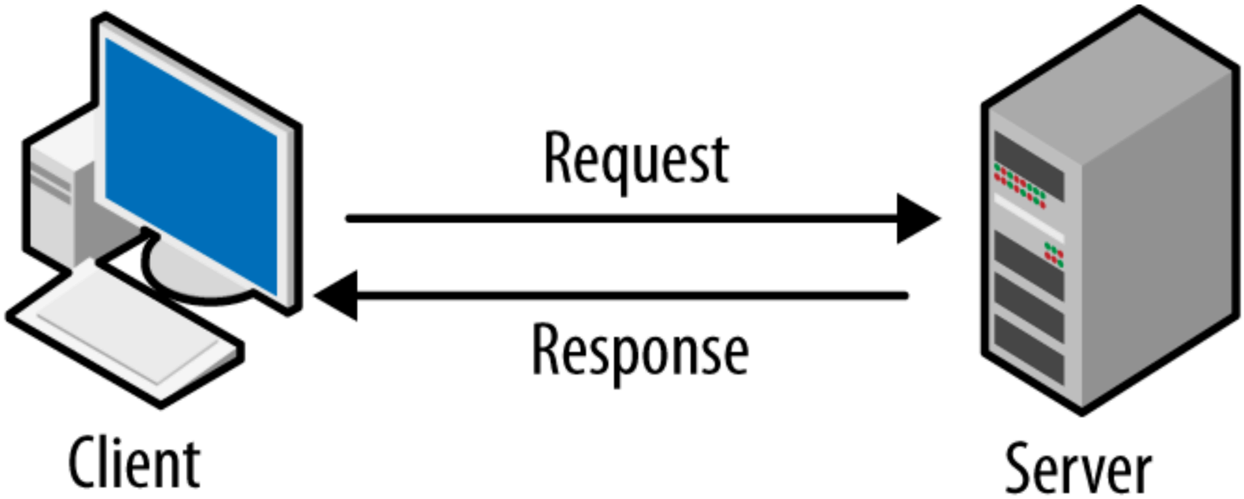
The **World Wide Web (WWW)** is a global collection of documents and other resources, linked by hyperlinks and URIs.

Internet Society (ISOC)

The **Internet Society (ISOC)**, founded in 1992, is dedicated to keeping the Internet open, transparent, and user-defined.

🧩 Simple Client-Server Model

A **client** (like a browser or app) sends a request to a **server**, which processes it and sends a response back.



📡 Network Protocols

TCP (Transmission Control Protocol)

Used for **reliable** and **ordered** delivery (e.g., web browsing, emails, file transfers).
Ensures no data loss.

UDP (User Datagram Protocol)

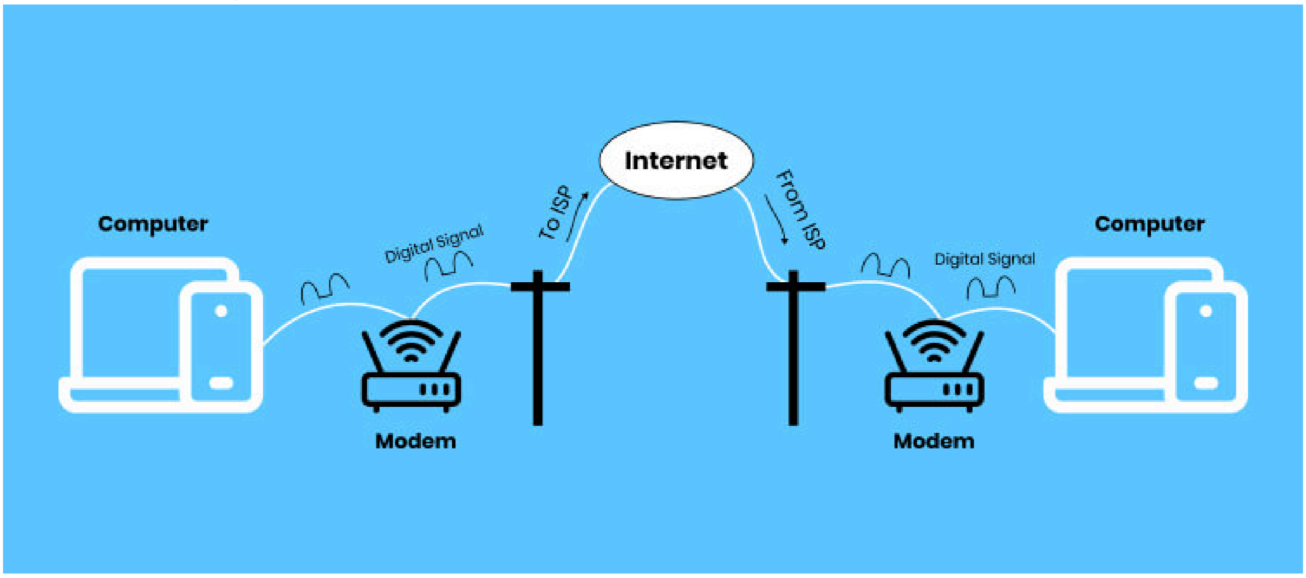
Used for **speed-critical**, real-time apps (e.g., gaming, streaming, video calls).
Faster but **unreliable**.

HTTP / HTTPS

- **HTTP**: Used for transferring non-sensitive web content.
- **HTTPS**: Secure version (uses SSL/TLS).
  Used for web communication between browsers and servers.

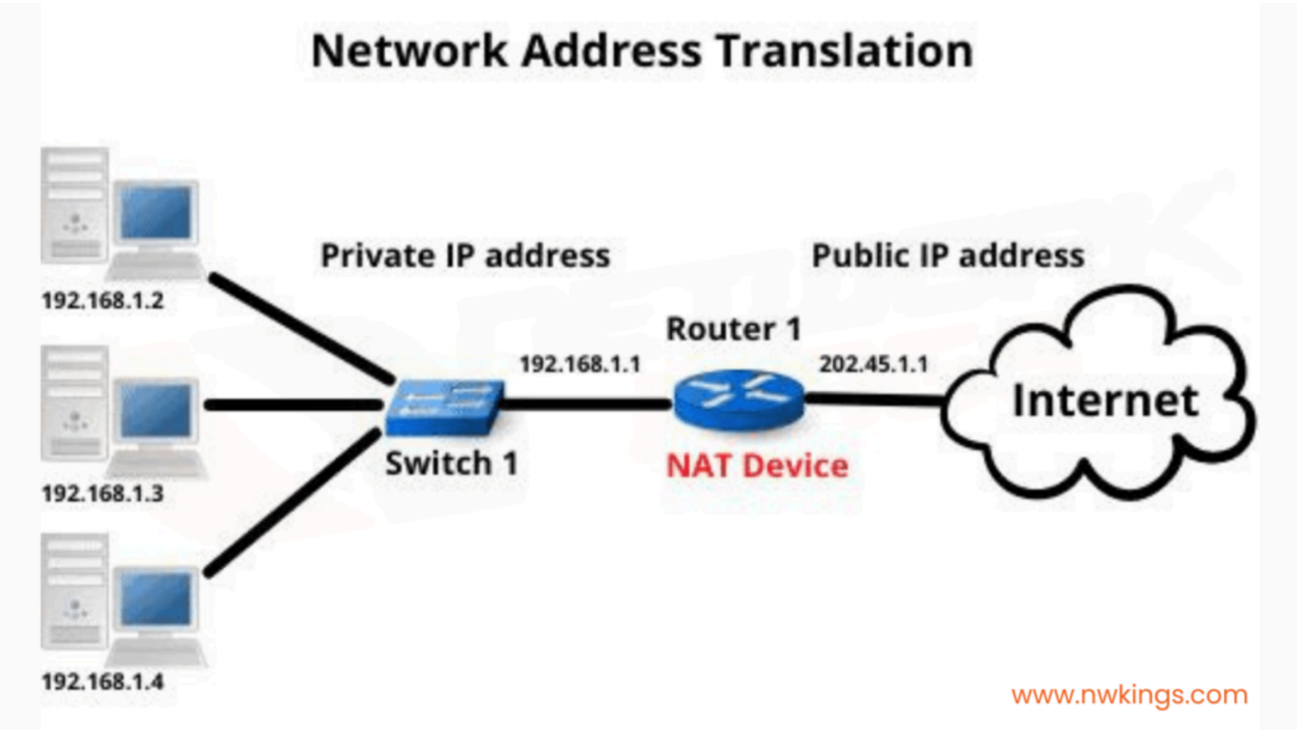
🌍 ISP (Internet Service Provider)

An **ISP** connects users to the internet.
Your device connects → ISP → Internet backbone → Data travels → Response returns via the **last mile** (ISP to your home).



NAT (Network Address Translation)

Allows multiple devices on a private network to share a single public IP address.
Adds a layer of security and conserves IPv4 addresses.



🧠 IP & Ports

- **IP Address** → Identifies the device.
- **Port Number** → Identifies the application.
- Total Ports: `2^16 ≈ 65,000` .

Common Ports

Service	Port
FTP	20/21
SSH	22
HTTP	80
HTTPS	443
Jenkins/Web Apps	8080 / 8443
Docker	2375 / 2376
K8s API	6443
Prometheus	9090
Elasticsearch	9200
Kibana	5601
MySQL	3306
PostgreSQL	5432

Types of Networks

Type	Description
LAN	Connects devices within a limited area (home/office).
MAN	Connects multiple LANs within a city/metropolitan area.
WAN	Connects networks over large geographic areas.

⚙️ Devices

Modem

Converts **digital** signals to **analog** (and vice versa) to connect local networks to the internet.

Router

Connects multiple devices to the same network and routes data between them and the internet.

🧱 Network Topologies

- **Bus**
- **Ring**
- **Star**
- **Tree**
- **Mesh**

🧬 OSI Model (7 Layers)

Sending a WhatsApp message "Hey!" from India to the US:

Layer	Name	Function	Data Unit	Example Protocols
7	**Application**	Message creation (WhatsApp App, HTTPS)	Message data	HTTPS, WebSocket
6	**Presentation**	Encryption & compression	Encrypted blob	AES, Curve25519
5	**Session**	Maintains persistent connection	Session stream	TLS, WebSocket
4	**Transport**	Reliable delivery, segmentation	TCP segments	TCP, UDP

Layer	Name	Function	Data Unit	Example Protocols
3	**Network**	Routing and addressing	Packets	IP, NAT, ICMP
2	**Data Link**	Framing, error detection	Frames	Wi-Fi, Ethernet
1	**Physical**	Bit transmission	Bits	Fiber, Wi-Fi

Example:

Your phone → Router → Internet → WhatsApp Server → Friend's phone
Each layer adds its own header (encapsulation).
On receiving, the reverse happens (de-encapsulation).

🔒 End-to-End Example Summary

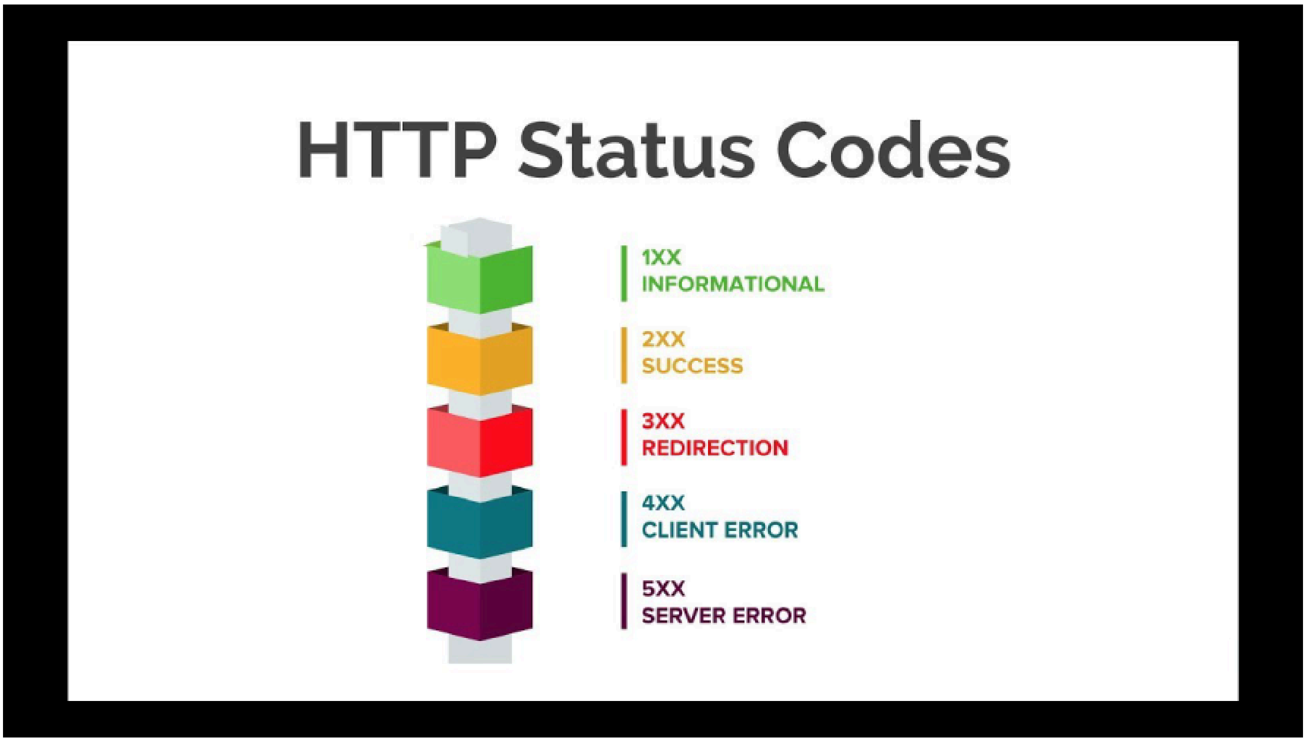
Component	Layer	Technology
Encryption	6/7	Signal Protocol
Reliable Transfer	4	TCP
Routing	3	IP, BGP
Local Transmission	2	Wi-Fi, LTE
Physical Medium	1	Fiber, 4G/5G

⚙️ TCP/IP Model

Simplified version of OSI with **5 layers** — combines application, presentation, and session into one.

🧩 MAC Address

A **MAC address** is a unique hardware ID assigned to a network card (Layer 2).
Used by **switches** to forward frames in a local network.



🍪 Cookies

A **cookie** is a small piece of data stored by the browser when you first log in — used for session management.

🌍 DNS (Domain Name System)

A **distributed database** that resolves domain names to IP addresses.
Lookup order:

1. Local DNS cache
2. ISP DNS
3. Root / Authoritative DNS

🤝 TCP 3-Way Handshake

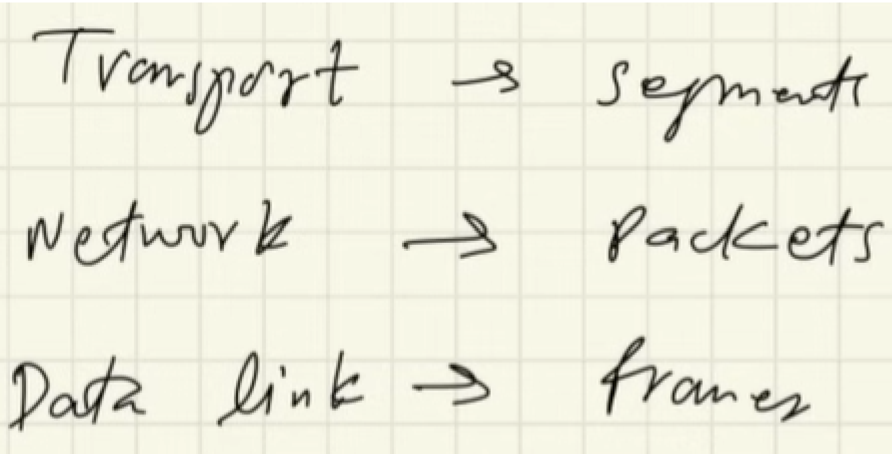
Step	Direction	Purpose	TCP State
1. SYN	Client → Server	Initiate connection	SYN_SENT
2. SYN-ACK	Server → Client	Acknowledge request	SYN_RECEIVED

Step	Direction	Purpose	TCP State
3. ACK	Client → Server	Confirm connection	ESTABLISHED

Analogy:
You: “Hello?” → Friend: “Hey, can you hear me?” → You: “Yes, loud and clear!” → Start chatting 🎉

🔁 Data Representation at Layers

- Layer 1 → **Bits**
- Layer 2 → **Frames**
- Layer 3 → **Packets**
- Layer 4 → **Segments**
- Layer 5–7 → **Data / Messages**



🧭 Loopback Address

Version	Address	Range	Description
IPv4	127.0.0.1	127.0.0.0/8	Localhost
IPv6	::1	::1/128	IPv6 localhost

Usage:
Used to test network functionality within your own computer (packets never leave the device).

🧠 How It Works When you send a packet to the loopback address (127.0.0.1): • The packet never leaves your computer. • It is handled entirely by the operating system's network stack. This helps you test local network functionality (like sockets, web servers, APIs) without needing an external network or internet.

Interview Tip:

> A loopback address (127.0.0.1) routes traffic internally to test local network functionality.

✅ **In Short:** When you send a WhatsApp message:

1. App encrypts and prepares data (Application layer).
2. Layers add addressing, routing, and framing info.
3. Data travels through routers and ISPs to the destination.
4. Receiver's device reverses the process and decrypts the message.

📘 Summary Table

Concept	Layer	Function
Encryption	6/7	Data security
Reliability	4	TCP delivery
Routing	3	IP addressing
Transmission	2	Frames
Signal	1	Bits

Networking

* **Ip address** :- Unique identification number assigned to device.
ipv4 address = eg. 172. 10. 68. 2
ipv4 address is made up of 4 bytes separated by .
1 byte = 8 bits ∴ ip is made of 32 bits.
each byte value varies from 0-255

> bit has only values :- 0, 1

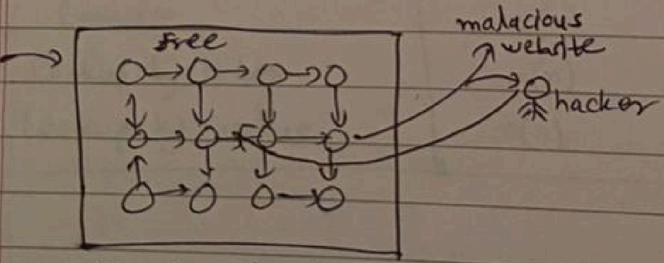
8 bits → 1 1 1 1 1 1 1 1

$2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$

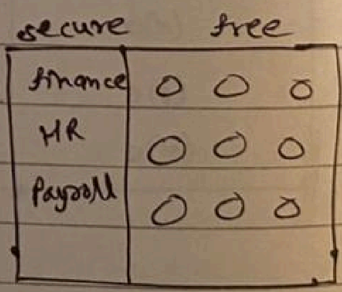
$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = \underline{255}$

* **Subnet** :- Let's take an example of office network. Everyone in office is using the same network. If someone accidently hits malacious website then hackers will get access to not only that device but all the office network. Hence we use concept of subnet which is nothing but dividing network (sub-networking) so that sensitive information should not be available easily to hacker.

office n/w →



Without subnet



subnetting

Advantages :-
→ security → Privacy → Isolation.

* **Types of subnet** :-
Private :- Does not have access to internet
Public :- has access to internet.

How to divide ip addresses according to subnet?

→ Using CIDR range.

eg.

let's say we have been provided ip range:- 172.16.0.0 to 172.16.255.255

Now, if we want 256 ip addresses to be secured then CIDR range will be :- 172.16.3.0/24 ⇒ any random ip from range followed by /24.

Calculating CIDR:-

We have total 4 Bytes (32 bits) in a ip address.

CIDR range formula :- 172.16.3.0/x

$$\boxed{\text{total ip addresses req.} = 2^{32-x}}$$

$\therefore 256 = 2^{32-x}$

$\therefore 2^8 = 2^{32-x} \quad \therefore 8 = 32-x \quad \boxed{\therefore x = 24}$

another example:-

let's say we want 32 ip addresses.

$\therefore 32 = 2^{32-x}$

$2^5 = 2^{32-x} \quad \therefore 32-x = 5 \quad \therefore \boxed{x = 27}$

$\boxed{\text{CIDR} = 172.16.4.0/27}$

Private IP address starts with:- 192. 172. 10.

Ports:- <ip address>:<port> eg. 10.221.145.10:80

port is nothing but a slot/gate which acts as entry/exit point of a device. Each application running on a device has unique port attached to it. User can access that application using this port.

IP address range -

Class	Address Range
A	10.0.0.0 – 10.255.255.255
B	172.16.0.0 – 172.31.255.255
C	192.168.0.0 – 192.168.255.255